"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totaly my own, I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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1) $\begin{bmatrix} 1 & \alpha & -y & \alpha \\ 3\alpha & 5 & -9y & 1 \\ 2 & 13 & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3\alpha & 2 & 1 \\ \alpha & 5 & 13 & 14 \\ -4 & -9y & 3 & -6 \\ \alpha & 1 & \alpha^2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 9\alpha^2+6 \\ 16\alpha + 28 + 14 \end{bmatrix}$

$9\alpha^2+6=1$

$9\alpha^2=-5$

$\alpha^2=\frac{-5}{9}$

$\alpha=\sqrt{\frac{5}{9}}$

$-y+27\alpha y$

2)

a) The column space and the null space of a matrix, where A is a m*n matrix then dimension Null(A) + dimension Column(A) = n. Here, dim Null (A) is number of free variables in row reduced form of A and a basis for col(A) is given by the columns corresponding to the leading 1's in the row

So, the dimension of the null space of a matrix is called the nullity of the matrix. The dimension of the column space of a matrix is called the rank of the matrix.

c) Yes, the solution space a vector space, Dimension of a vector space: The dimension of a finite dimensional vector space is denoted by dim(v) is defined to be the number of vectors in a basis for v. In addition we define the zero vector space to have dimension zero

Finite dimensional: A non zero vector space v is called finite dimensional if it contains a finite set of vector form a basis. If no such set exist v is called infinite dimensional.

b)

3) $H = I - 2uu^T$

a)

$$H^T = (I - 2uu^T)^T$$
$$= I^T - 2(uu^T)^T$$
$$= I - 2(u^T)^T u^T$$
$$= I - 2uu^T$$

$H^T = H$ → symetric

$$H^T H = H^2 = (I - 2uu^T)(I - 2uu^T)$$
$$H^2 = I - 4uu^T + 4uu^T(uu^T)$$
$$H^2 = I - 4uu^T + 4u(u^T u)u^T$$
$$H^2 = I - 4uu^T + 4uu^T$$
$$H^2 = I$$

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b) $u^T u = \|u\|^2 = 1$ so

$H^T H = HH$

$= (I - 2uu^T)(I - 2uu^T)$

$= I - 2uu^T - 2uu^T + 4(uu^T)(uu^T)$

$= I - 4uu^T + 4u(u^T u)u^T$ ($\|u\| = 1$)

$= I$

(same calculations as (a) part)

similarly, $H^T H = I$

so, $HH^T = H^T H = I$

so, H is an orthogonal matrix

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c) For a matrix A, a vector (non zero) will be eigen vector of A corresponding eigen value λ if, $Av = \lambda v$

Now, $Hu = (I - 2uu^T)u$

$= Iu - 2uu^T u$

$= u - 2u.\|u\|^2$

matrix, so I is the only eigenvalue of I and, $Iu = I.u = u$

and $\|u\| = +\sqrt{u^T u}$ u is a unit vector, so, $\|u\| = 1 = +\sqrt{u^T u}$

and hence, $u^T u = \|u\|^2 = I$

c) hence, $Hu = u - 2u \cdot 1 = -u$

$\Rightarrow Hu = -u = (-1)u$

so, u is an eigen vector of H and corresponding eigen value for u is $\lambda = -1$

d) Given, $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2(u^T)^T u^T$

$= I - 2uu^T = H$

so, H is a symmetric matrix

and there is a theorem which states that,

$H = Q \Lambda Q^T$ (Where, Q is an orthogonal matrix)

$= Q \Lambda Q^{-1}$ (As, $QQ^T = I \Rightarrow Q^T = Q^{-1}$)

$= (Q^{-1})^{-1} \Lambda Q^{-1}$ (Λ = diagonal matrix comits of eigen values of H)

$= P^{-1} \Lambda P$

so it is diagnolizable matrix